

Kyiv Mechmat 2025

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February 2025

1 Problems

P 4. Let f be continuous on $[a, b]$ and differentiable and increasing on (a, b) . Show that there exists $c \in (a, b)$ such that

$$f'(c) = \sqrt{\frac{(f(c) - f(a))(f(b) - f(c))}{(c - a)(b - c)}}.$$

Define $g(x) = \frac{2f(x) - f(a) - f(b)}{f(b) - f(a)}$ and $h(x) = \frac{2x - a - b}{b - a}$. Note that g, h are increasing on (a, b) and $g(a) = h(a) = -1$ and $g(b) = h(b) = 1$. Hence, $G(x) = \arcsin g(x)$ and $H(x) = \arcsin h(x)$ are well defined on $[a, b]$ and in particular, continuous. G, H are also differentiable.

By Rolle's on the function $(G - H)(x)$, there exists $c \in (a, b)$ such that $G'(c) = H'(c)$,

$$\begin{aligned} \frac{\frac{2f'(c)}{f(b) - f(a)}}{\sqrt{1 - \left(\frac{2f(c) - f(a) - f(b)}{f(b) - f(a)}\right)^2}} &= \frac{\frac{2}{b - a}}{\sqrt{1 - \left(\frac{2c - a - b}{b - a}\right)^2}} \\ \frac{2f'(c)}{\sqrt{(f(b) - f(a))^2 - (2f(c) - f(a) - f(b))^2}} &= \frac{2}{\sqrt{(b - a)^2 - (2c - a - b)^2}} \\ \frac{f'(c)}{\sqrt{(f(c) - f(a))(f(b) - f(c))}} &= \frac{1}{\sqrt{(c - a)(b - c)}}. \end{aligned}$$

Note that the denominator was simplified by difference of squares. Rearranging gives the desired equality.

P 5. Let matrices $A, B \in M_n(\mathbb{R})$ such that $A^2 + 2BA = 3B - A + I$ and $2B^2 + AB = 3A - B - I$. Prove that $\det(A - B) > 0$.

Note that

$$\begin{aligned}
(A + 2B - I)^2 &= A^2 + 2AB - A + 2BA + 4B^2 - 2B - A - 2B + I \\
&= (A^2 + 2BA) + 2(2B^2 + AB) - 2A - 4B + I \\
&= (3B - A + I) + 2(3A - B - I) - 2A - 4B + I \\
&= 3(A - B).
\end{aligned}$$

Hence, $\det(A - B) = 3^n \det(A + 2B - I)^2 \geq 0$. It remains to show $\det(A + 2B - I) \neq 0$. Suppose FTSOC there exists a nonzero vector x such that $(A + 2B - I)x = 0$. Then $(A - B)x = 0$, so $Ax = \frac{x}{3}$ and $Bx = \frac{x}{3}$. Now $(A^2 + 2BA)x = x/3$ but $(3B - A + I)x = 5x/3$, a contradiction.